MATERIAL SAFETY DATA SHEET

GHS / OSHA HazCom Compliant Safety Data Sheet

KETONE LUBRICANTS

Product Name:	CUTLIGHT VC 212	**Article No.:**	1000600
Chemical Type:	Water Soluble Cutting Oil	**Batch No.:**	LCT - 0723
Manufacturer:	KETONE Specialty Lubricants	**Production Date:**	07-2023
Net Container:	4 Ltrs (Available 20L / 24L / 208L)	**Expiry Date:**	07-2025

1. IDENTIFICATION OF THE SUBSTANCE / PREPARATION AND COMPANY

Product Name	CUTLIGHT VC 212 Water Soluble Cutting Oil
Identified Uses	Industrial metalworking fluid, soluble cutting oil for machining, grinding, and metal processing operations.
Restrictions on Use	For Industrial Use Only. Not for domestic or consumer application.
Manufacturer / Supplier	KETONE Specialty Lubricants & Chemicals

2. HAZARDS IDENTIFICATION

GHS CLASSIFICATION & HAZARD STATEMENTS

Classification: Skin Irritation Category 2; Eye Irritation Category 2A; Skin Sensitization Category 1; Aquatic Hazard (Chronic) Category 3.

- **H315:** Causes skin irritation.
- **H317:** May cause an allergic skin reaction.
- **H319:** Causes serious eye irritation.
- **H412:** Harmful to aquatic life with long lasting effects.

Precautionary Statements:

- **P261:** Avoid breathing mist / vapours / spray.
- **P280:** Wear protective gloves, protective clothing, eye protection, and face protection.
- **P273:** Avoid release to the environment.
- **P302+P352:** IF ON SKIN: Wash with plenty of water and soap.
- **P305+P351+P338:** IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing.

3. COMPOSITION / INFORMATION ON INGREDIENTS

Chemical Name / Component	CAS Number	Weight %	Classification
Highly Refined Mineral Base Oils (C15 - C50)	64742-54-7 / Mixture	60 - 80 %	Not classified as hazardous
Emulsifiers & Corrosion Inhibitor Package	Proprietary Mixture	15 - 25 %	Skin Irrit. 2 (H315), Eye Irrit. 2 (H319)
Biocides / Anti-microbial Additives	Proprietary Mixture	1 - 5 %	Skin Sens. 1 (H317), Aquatic Chronic 3 (H412)
Coupling Agents & Secondary Alkanolamines	102-71-6 / Proprietary	1 - 5 %	Eye Irrit. 2A (H319)

4. FIRST AID MEASURES

- **In Case of Contact (Eye):** Flush eyes immediately with plenty of water for at least 15 minutes while lifting lower and upper eyelids. Remove contact lenses if present. Get medical attention if irritation persists.
- **In Case of Contact (Skin):** Immediately remove contaminated clothing and shoes. Wash affected area thoroughly with soap and plenty of water. Wash clothing before reuse. If irritation or rash develops, seek medical attention.
- **If Inhaled:** Remove victim to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. In all cases where breathing is difficult, obtain immediate medical attention.
- **If Swallowed:** DO NOT INDUCE VOMITING unless directed by medical personnel. Never give anything by mouth to an unconscious person. Give large quantities of water if conscious. Seek immediate medical attention.

5. FIRE FIGHTING MEASURES

- **Extinguishing Media:** Carbon dioxide (CO2), dry chemical powder, foam, or water fog. Do not use direct heavy water jet as it may scatter the burning liquid.
- **Specific Hazards:** Water soluble emulsion can boil or foam when exposed to high heat. Thermal decomposition product gases may include carbon oxides (CO, CO2), nitrogen oxides (NOx), sulfur oxides, and smoke.
- **Protective Equipment for Firefighters:** Wear self-contained breathing apparatus (SCBA) operated in positive pressure mode and full protective gear.

6. ACCIDENTAL RELEASE MEASURES

- **Personal Precautions:** Wear suitable protective clothing, chemical resistant gloves, and eye/face protection. Ensure adequate ventilation. Floors will be slippery; execute caution to prevent falls.
- **Environmental Precautions:** Prevent material from entering drains, surface water, groundwater, or soil containment system.
- **Clean-up Methods:** Absorb liquid spillage with inert absorbent material (sand, diatomaceous earth, vermiculite, saw dust). Sweep/shovel into clean, appropriately labelled disposal containers. Clean contaminated surfaces with detergent and water.

7. HANDLING AND STORAGE

- **Handling Precautions:** Do not get in eyes, on skin, or on clothing. Do not breathe vapor or mist. Use only with adequate local exhaust ventilation. Wash thoroughly after handling. Remove and wash contaminated clothing before reuse. Keep container closed when not in use.
- **Storage Conditions:** Store in original, tightly closed container in a dry, well-ventilated location. Recommended storage temperature range: **5°C to 40°C**. Protect against direct sunlight and extreme temperatures (avoid freezing and heat above 50°C). Keep away from strong oxidizing agents.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limit	Oil mist (mineral): ACGIH TLV-TWA: 5 mg/m³; STEL: 10 mg/m³ (8-hour shift).
Engineering Controls	Use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits.
Eye Protection	Safety goggles with side shields or full face shield where splashing risk exists.
Hand / Skin Protection	Chemical-resistant impervious gloves (Nitrile rubber, Neoprene). Protective apron and long sleeves as needed.
Respiratory Protection	Approved oil-mist respirator or organic vapor cartridge respirator if concentration exceeds workplace exposure limits.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance / State	Liquid (Blue Container)	**Solubility in Water**	Soluble (Forms Emulsion)
Odor	Mild hydrocarbon / characteristic	**pH (5% Emulsion)**	8.5 – 9.5

Kinematic Viscosity	32 – 46 mm²/s @ 40°C	**Flash Point (COC)**	> 160°C (320°F)
Specific Gravity / Density	0.88 – 0.92 g/cm³ @ 15°C	**Boiling Point**	> 200°C

10. STABILITY AND REACTIVITY

- **Chemical Stability:** Stable under recommended storage conditions (5°C - 40°C).
- **Conditions to Avoid:** Direct heat, open flames, sparks, static discharge, and freezing temperatures. Extended high-temperature exposure.
- **Incompatible Materials:** Strong oxidizing agents, strong acids, and strong reducing agents.
- **Hazardous Decomposition Products:** Carbon monoxide, carbon dioxide, nitrogen oxides, sulfur compounds, and toxic fumes under combustion.

11. TOXICOLOGICAL INFORMATION

- **Acute Toxicity:** Oral LD50 (rat) > 5000 mg/kg; Dermal LD50 (rabbit) > 2000 mg/kg (Estimated based on components).
- **Skin Contact:** Causes mild to moderate skin irritation. Prolonged or repeated contact may cause skin defatting, dermatitis, or allergic reaction.
- **Eye Contact:** Direct splash causes moderate to severe eye irritation, redness, and tearing.
- **Inhalation:** Inhalation of mists or vapors generated at high temperatures may cause respiratory tract irritation, coughing, or nausea.
- **Carcinogenicity:** Mineral base oils are highly refined and not classified as carcinogenic by IARC, NTP, or OSHA.

12. ECOLOGICAL INFORMATION

- **Ecotoxicity:** Harmful to aquatic organisms. May cause long-term adverse effects in the aquatic environment.
- **Persistence and Degradability:** Base oil components are inherently biodegradable; emulsifier components biodegrade in water emulsion.
- **Bioaccumulative Potential:** Bioaccumulation potential is low to moderate.

13. DISPOSAL CONSIDERATIONS

- **Waste Disposal:** Dispose of contents and container in accordance with local, state, national, and international environmental regulations. Do not pour into drains, surface water, or soil. Used cutting fluid emulsions should be processed by an authorized waste management facility.

14. TRANSPORT INFORMATION

- **ADR / RID / IMDG / ICAO / IATA:** Not dangerous goods / Not classified as hazardous for transport under international transport regulations. Keep container upright during transport.

15. REGULATORY INFORMATION

- **GHS Labeling:** Compliant with Globally Harmonized System of Classification and Labelling of Chemicals (GHS).
- **Occupational Health & Safety:** Ensure compliance with national occupational health, chemical safety, and industrial safety standards.

16. OTHER INFORMATION

Disclaimer: The information contained in this Material Safety Data Sheet (MSDS) is based on the present state of knowledge and accurate to the best of our knowledge. It describes the product with regard to appropriate safety precautions and does not represent a guarantee of the properties of the product. KETONE and its distributor accept no liability for damage or injury resulting from misuse or failure to adhere to recommended storage and handling procedures.

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